

PROJECT EXPO 2026

Smart IV Drip Monitoring Using IOT

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PROBLEM STATEMENT:

Intravenous (IV) fluid therapy is one of the most commonly used medical procedures in hospitals, requiring continuous monitoring to ensure accurate and safe fluid administration. In many healthcare settings, nurses manually inspect IV drip bottles and regulate flow rates at regular intervals. This manual approach is time-consuming, prone to human error, and becomes increasingly challenging when a single healthcare provider is responsible for multiple patients.

Failure to detect changes in the IV flow rate or an empty IV bottle in time can lead to serious complications such as under-infusion, over-infusion, reverse blood flow, air embolism, and interruption of treatment. Existing monitoring methods often lack real-time remote access and automated intervention, resulting in delayed responses to critical situations.

Therefore, there is a need for an intelligent, low-cost, and IoT-enabled IV drip monitoring system that can continuously monitor drip rate and fluid level, transmit real-time data to healthcare providers, generate timely alerts for abnormal conditions, and automatically regulate or stop the IV flow when necessary. Such a system can improve patient safety, reduce the workload of medical staff, minimize human errors, and enhance the efficiency of hospital-based IV therapy management.

ABSTRACT:

Manual monitoring of intravenous (IV) drip therapy remains a routine but risk-prone task in hospitals, where nurses must periodically check bottle levels and flow rates across multiple patients. Delays in detection can lead to serious complications such as reverse blood flow, air embolism, or incorrect dosage delivery.

This project proposes a Smart IV Drip Monitoring System using an ESP32 microcontroller integrated with IoT technology to provide real-time and remote monitoring. A real IV infusion set is used, and an IR (infrared) sensor is positioned at the drip chamber to detect and count individual drops. Based on the drip count, the system calculates the flow rate in ml/min, enabling accurate monitoring of fluid delivery.

Instead of relying on local displays alone, the system utilizes the built-in Wi-Fi capability of ESP32 to transmit real-time data to a web-based dashboard or mobile application. This allows healthcare providers to monitor multiple patients remotely and respond quickly to abnormal conditions.

When the flow rate deviates from the prescribed range or when the IV bottle approaches empty, the system can trigger alerts and initiate automatic corrective action using a servo motor to regulate or stop the flow. Additionally, safety considerations such as prevention of reverse blood flow using anti-reflux mechanisms are incorporated in the system design.

By combining real-time sensing, automated control, and IoT-based remote monitoring in a low-cost and scalable solution, the proposed system enhances patient safety, reduces manual workload, and improves response time in hospital environments.

OBJECTIVES:

The primary objective of this project is to design and develop a Smart IV Drip Monitoring System using an ESP32 microcontroller integrated with IoT technology to provide continuous and real-time monitoring of intravenous fluid administration.

The specific objectives are:

- ◆ To continuously monitor the IV drip flow using an IR sensor placed at the drip chamber.
- ◆ To accurately detect and count individual drops and calculate the corresponding flow rate in ml/min.
- ◆ To monitor the IV bottle condition and identify abnormal situations such as low fluid level, empty bottle, or abnormal drip rate.
- ◆ To transmit real-time monitoring data to a web-based dashboard or mobile application using the built-in Wi-Fi capability of the ESP32.
- ◆ To generate immediate visual and audible alerts whenever abnormal conditions are detected.
- ◆ To automatically regulate or stop the IV fluid flow using a servo motor to prevent complications caused by incorrect flow or an empty IV bottle.

- ◆ To incorporate anti-reflux safety mechanisms to minimize the possibility of reverse blood flow.
- ◆ To reduce the dependency on continuous manual monitoring by healthcare staff while improving patient safety and treatment reliability.
- ◆ To develop a low-cost, compact, and scalable prototype that can be easily implemented in hospitals and healthcare centers.

SCOPE OF THE PROJECT:

This project focuses on the design, development, and testing of a Smart IV Drip Monitoring System based on the ESP32 microcontroller and IoT technology. The system uses an IR sensor to detect individual IV drops in the drip chamber and calculate the flow rate in real time. The ESP32 processes the sensor data and transmits it through Wi-Fi to a web dashboard or mobile application, allowing healthcare professionals to monitor patients remotely.

The project also includes the implementation of a servo motor that automatically regulates or stops the IV fluid flow whenever abnormal conditions such as incorrect drip rate or a nearly empty IV bottle are detected. Safety is further enhanced by incorporating an anti-reflux mechanism to reduce the risk of reverse blood flow.

The prototype is intended to work with a standard IV infusion set without requiring major modifications to existing hospital equipment. It is designed as a low-cost and scalable solution suitable for hospitals, clinics, and healthcare centers. The project scope is limited to prototype development, laboratory testing, and performance evaluation. Clinical validation, hospital-wide deployment, integration with electronic health record systems, and regulatory certification are considered future enhancements and are beyond the scope of this project.

EXPECTED OUTCOMES:

The expected outcome of this project is the successful development of a functional Smart IV Drip Monitoring System capable of continuously monitoring IV fluid administration with improved accuracy and reliability. The system is expected to detect the drip rate in real time using an IR sensor, calculate the corresponding flow rate, and identify abnormal conditions such as an empty IV bottle, low fluid level, or incorrect drip rate.

Whenever an abnormal condition is detected, the system will generate immediate alerts through the IoT platform and automatically activate the servo motor to regulate or stop the IV fluid flow, thereby reducing the possibility of reverse blood flow, air embolism, and incorrect dosage delivery.

The ESP32-based IoT platform will enable healthcare providers to remotely monitor multiple patients through a web dashboard or mobile application, improving response time and reducing the need for continuous bedside supervision. The completed prototype is expected to demonstrate a low-cost, compact, energy-efficient, and scalable solution that enhances patient safety, reduces nurses' workload, improves monitoring efficiency, and provides a strong foundation for future smart hospital applications such as multi-bed monitoring, cloud data storage, and predictive healthcare analytics.

LITERATURE SURVEY:

The following table surveys ten representative studies spanning IoT-based IV/saline drip monitoring, load-cell and optical sensing approaches, servo/actuator-based flow control, and cloud/app-connected alerting, situating this project within the current state of the art.

Author	Contribution	Limitation
Nandhini et al. (2024)	IoT-based drip monitoring using load-cell amplifier with GSM/SMS alerting to nursing staff.	Alerting only through SMS; no automated flow correction mechanism.
Preethi et al. (IEEE, 2020)	Combines IoT healthcare monitoring with intravenous flow-control features.	Depends on continuous Wi-Fi connectivity; single-bed prototype only.
Anagha et al. (IJRET, 2020)	IoT-based intravenous flow monitoring system reporting drip status to staff.	No automatic shut-off or corrective actuation on abnormal flow.

Joseph et al. (IEEE, 2019)	Drip monitoring system for smart hospitals using periodic level sensing.	Notification pipeline relies on GSM/SMS, introducing alert delay and cost.
IEEE 10053449 (2023)	Smart IV system with reverse-blood-flow prevention and electrolyte-level checks.	Multi-sensor electrolyte assessment adds hardware complexity and cost.
IEEE 9297376 (2020)	IoT-enabled infusion-rate monitoring device using an IR drop sensor and Wi-Fi module.	Actuation is keypad/app-driven rather than autonomous; single-bed scope.
IEEE 9848118 (2022)	Smart IV drip monitor with a dedicated air-bubble detection indicator.	Focused narrowly on bubble detection; lacks flow-rate correction.
MediDrip, Zenodo preprint (2026)	ESP32 + HX711 load-cell system with Blynk app integration for remote volume tracking.	Weight-based sensing is sensitive to bottle placement and needs recalibration.
DripControl+ / TDR study (academia.edu)	Microwave TDR and optocoupler sensing with servo/peristaltic actuation and app control.	TDR electronics raise cost and power consumption; not field-portable.
IJARCCE Saline Level System (2019)	NodeMCU-based system using a micro-servo to auto-stop flow at an empty bottle.	Local buzzer alert only; no IoT app connectivity for remote staff visibility.

Anand et al. (Thamaraikani, IIJSR 2018)	Intravenous drip monitoring system with GSM-based low-level alerting to nursing staff.	Early-stage system with no automated actuation on the drip line.
Bhavasaar et al. (IEEE TIAR, 2016)	Automated intravenous fluid monitoring and alerting system using sensor-based level detection.	Limited to buzzer/LED alerts on-site; no mobile app or remote visibility.

Across the surveyed literature, three broad approaches recur: GSM/SMS-based alerting, load-cell or optical volume sensing, and servo/actuator-based manual flow correction. Very few of the surveyed works combine autonomous corrective actuation with dual local-and-remote alerting in a single low-cost unit, which is precisely the gap this project's design targets.

EXISTING SOLUTION ANALYSIS:

A range of existing approaches address parts of the IV-monitoring problem — from purely manual observation to specialised load-cell, optical, and microwave-based sensing systems, some paired with GSM alerts, cloud dashboards, or actuator-based flow control. The table below compares these approaches against their working principle, advantages, and limitations.

S. No	Existing Solution	Technology Used	Working Principle	Advantages	Limitations
1	Manual Visual Monitoring	Human observation	Nurse performs periodic bedside rounds to check bottle level	No cost, simple, no hardware	Error-prone, delayed, unscalable with high patient load

2	GSM/SMS Alert Systems	GSM module + level sensor	Sends SMS to nurse when fluid reaches a low threshold	Works without internet	No automatic flow correction; SMS delay and recurring cost
3	Load-Cell Based Systems	HX711 + load cell	Weighs the bottle to estimate remaining volume	Accurate volume estimation	Sensitive to bottle placement/vibration; needs recalibration
4	IR / Optical Drop Counters	IR photo-interrupter	Counts drops passing through the drip chamber for flow rate	Low cost, simple, fast response	No direct volume reading; affected by ambient light/positioning
5	Bubble / Air Detection Systems	IR or ultrasonic sensor	Detects air bubbles entering the IV line	Improves patient safety against air embolism	Limited to bubble detection; no flow-rate awareness
6	Cloud Dashboard Systems	ESP32/NodeMCU + Blynk/AWS	Streams live status to a remote dashboard or app	Real-time remote visibility across beds	Depends on internet/cloud uptime; no local mechanical safeguard

7	Microwave TDR Systems	Time-domain reflectometry sensor	Measures fluid level via reflected microwave signal	Precise, non-contact sensing	Expensive, power-hungry, complex electronics
8	Servo/Peristaltic Actuator Systems	Servo motor + keypad/app	Operator sets desired drip rate; servo pinches the tube	Active flow correction is possible	Correction is manually triggered, not autonomous
9	Multi-Bed Hospital IoT Platforms	Raspberry Pi + AWS IoT Core	Centralized dashboard monitoring many beds at once	Scales across an entire ward	High infrastructure cost and complex deployment
10	Noncontact Capacitive Monitors	Capacitive liquid-level sensor	Detects fluid presence without touching the fluid path	Sterile, no tubing modification needed	Comparatively new; limited large-scale clinical validation
11	Bluetooth Keypad-Controlled Systems	Bluetooth module + keypad/app	Staff manually sets drip rate via a nearby Bluetooth-linked app	No internet dependency; allows manual override	Short range; staff must be physically close to the bed

12	RFID-Linked Patient Monitoring	RFID tag + sensor	Links drip data to a specific patient ID for record-keeping	Reduces patient/bottle mix-up errors	Adds infrastructure; not focused on flow correction itself

Proposed Solution Overview: Overcoming the Gaps in Existing Systems

Building on the limitations identified above, the Smart IV Drip Monitoring system is designed specifically to close the gap between an abnormality occurring and it being both noticed and corrected. The table below maps each recurring limitation in existing systems to the corresponding response built into the proposed solution.

Limitation in Existing Systems	How the Proposed System Responds
Depends entirely on manual nurse visits to notice an abnormal drip.	Continuous, automated IR-based flow sensing at the drip chamber, 24/7, without needing a staff member present.
Alerts, where present, mostly rely on GSM/SMS with cost and delivery delay.	Instant dual alerting: an on-device LCD for anyone nearby, plus a Wi-Fi/IoT push notification to a mobile app for remote staff.

Correction of an abnormal flow requires a nurse to walk over and manually adjust the line.	A servo motor automatically clamps or releases the IV line the moment an abnormality is detected, correcting the condition before it becomes dangerous.
Weight/load-cell designs are sensitive to bottle placement and need recalibration.	Direct optical drop counting at the chamber is less sensitive to how the bottle is hung, giving a more stable flow-rate reading.
Precision sensing options (TDR, multi-sensor electrolyte systems) are costly and power-hungry.	Built from a single ESP32 board, an IR sensor, and a small servo — low cost, low power, easy to replicate across many beds.

GAP ANALYSIS:

Comparing the surveyed literature and existing solutions against the requirements of a practical, ward-deployable IV safety system reveals several recurring gaps, which this project specifically targets.

1. Delayed, SMS-Only Alerting

Limitation observed: Systems such as Nandhini et al. and Joseph et al. route every alert through GSM/SMS, which introduces network delay, per-message cost, and a single point of dependency on the cellular network.

How the Smart IV Drip Monitoring system addresses it: The system pushes alerts simultaneously to a bedside LCD (local, instant) and to a mobile app over Wi-Fi/IoT (remote), so staff are notified through two independent channels without relying on SMS delivery.

2. No Autonomous Corrective Action

Limitation observed: Most surveyed systems, including the IR-sensor monitor of Preethi et al. and the flow-monitoring work of Anagha et al., only raise an alert and still require a nurse to physically intervene to correct the flow.

How the Smart IV Drip Monitoring system addresses it: A servo motor is directly coupled to the drip-line clamp so that, the moment an abnormal flow condition is detected, the system pinches or releases the tube automatically — reducing dependence on immediate human response.

3. Placement-Sensitive Load-Cell Sensing

Limitation observed: Weight-based designs such as MediDrip rely on a load cell under the bottle; small disturbances, bottle-hanger vibration, or incorrect mounting can skew readings and require frequent recalibration.

How the Smart IV Drip Monitoring system addresses it: Flow is measured directly at the drip chamber using an IR drop sensor, which counts individual drops rather than inferring volume from weight, making the reading far less sensitive to how the bottle is hung.

4. High Cost and Complex Hardware

Limitation observed: Approaches built around microwave TDR sensing or multi-sensor electrolyte assessment (e.g., the DripControl+ study and IEEE 10053449) achieve precision but at a hardware and power cost that is hard to justify for routine ward deployment.

How the Smart IV Drip Monitoring system addresses it: The design uses a single low-cost ESP32 board, an IR sensor, and a small servo motor — off-the-shelf components that keep the bill of materials low enough for deployment across many beds in a ward.

5. Single-Parameter, Single-Channel Monitoring

Limitation observed: Several systems specialise in one concern only — bubble detection (IEEE 9848118), flow counting, or manual keypad-driven rate control — without combining detection, correction, and dual-channel alerting in one device.

How the Smart IV Drip Monitoring system addresses it: The system unifies flow-rate sensing, automatic servo-based correction, an on-device LCD display, and IoT mobile-app notification into a single compact unit, closing the gap between detecting a problem and acting on it.

6. Correction That Still Depends on Staff Proximity

Limitation observed: Bluetooth-keypad and app-driven servo systems can correct flow, but only after a staff member is close enough to issue the command manually, which reintroduces the same response delay the automation was meant to remove.

How the Smart IV Drip Monitoring system addresses it: Correction logic runs locally on the ESP32 itself, so the servo reacts to sensor readings the instant a threshold is crossed, independent of whether any staff member is nearby or connected to the app at that moment.

7. Fixed, Non-Configurable Alert Thresholds

Limitation observed: Many of the surveyed low-cost designs hard-code a single low-level or rate threshold in firmware, which does not suit the wide range of prescribed drip rates used across different patients and medications.

How the Smart IV Drip Monitoring system addresses it: The mobile app allows the assigned nurse to set the expected flow-rate range for each patient, so the same hardware can be reused correctly across patients with very different prescribed infusion rates.

Integration Gap: Lack of a Combined Detect-and-Correct Platform

Most surveyed systems focus on only one capability — sensing, alerting, or actuation — rather than integrating all of them into a single affordable device. The Smart IV Drip Monitoring system combines IR-based flow sensing, autonomous servo-driven flow correction, a local LCD status display, and IoT-based mobile-app alerting into one compact embedded platform, reducing both the risk window between detection and correction and the deployment complexity for hospital staff.

SIGNIFICANCE OF THE PROJECT:

The Smart IV Drip Monitoring System using IoT is designed to improve the safety, efficiency, and reliability of intravenous (IV) therapy in hospitals. Since manual monitoring of IV drips is still common in many healthcare facilities, there is always a risk of delayed detection of problems such as an empty IV bottle, incorrect flow rate, reverse blood flow, or air embolism. These conditions can seriously affect patient health if not identified and corrected immediately.

The proposed system continuously monitors the IV drip flow using an IR sensor and calculates the flow rate in real time. By integrating the ESP32 microcontroller with IoT technology, the system transmits live monitoring data to a web dashboard or mobile application, enabling healthcare providers to supervise multiple patients remotely without being physically present at each bedside. This significantly reduces the workload of nurses while improving the efficiency of patient care.

An important feature of the project is its ability to provide automatic corrective action. Whenever the system detects an abnormal flow rate or a nearly empty IV bottle, it immediately generates alerts and activates a servo motor to regulate or stop the IV flow. This reduces the response time, minimizes human error, and helps prevent serious medical complications. The inclusion of an anti-reflux mechanism further enhances patient safety by reducing the possibility of reverse blood flow.

Another major significance of this project is its affordability and scalability. The system is developed using low-cost components such as the ESP32, IR sensor, and servo motor, making it suitable for deployment in hospitals, clinics, and rural healthcare centers with limited resources. The prototype can also be easily upgraded in the future to support

multi-patient monitoring, cloud-based data storage, predictive analytics, and integration with smart hospital management systems.

Overall, this project contributes to the advancement of smart healthcare by combining real-time sensing, automated control, and IoT-based remote monitoring into a single embedded system. It enhances patient safety, reduces manual monitoring, improves healthcare efficiency, and supports the future development of intelligent hospital infrastructure.

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